Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3 Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

1 Local Food Market Stall [C1-1 C2-2]

A local food market company is planning its stall offering for an upcoming market period. A standard stall is expected to generate profits of eighteen thousand US dollars under high demand and eleven thousand US dollars under low demand, while a deluxe stall is expected to generate thirty-one thousand US dollars and six thousand US dollars, respectively. The company must select its stall before the actual level of customer demand is known.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Standard stall	A_2: Deluxe stall
States of nature	S_1: High demand	S_2: Low demand
Uncertain event	Actual market demand after the stall decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. Monetary measure: Profit in thousand US dollars for each alternative–state pair	

C2 - Payoff table [2]

Each supplied monetary payoff is placed at the intersection of its alternative and state:

$$P(A_1, S_1) = 18, \quad P(A_1, S_2) = 11,$$
$$P(A_2, S_1) = 31, \quad P(A_2, S_2) = 6.$$

Thus, in thousand US dollars, the completed payoff table is

Alternative	High S_1	Low S_2
Standard stall A_1	18	11
Deluxe stall A_2	31	6

2 Small Bakery Production [C1-1 C2-3]

A small bakery is planning its production capacity for an upcoming sales period. A standard oven can generate 3 thousand BRL per production batch, while an industrial oven can generate 5 thousand BRL per batch. Based on expected demand,

the standard oven would complete 16 batches under high demand or 11 under low demand, whereas the industrial oven would complete 13 or 9 batches, respectively. In addition, the bakery would receive a fixed sales bonus of 6 thousand BRL if demand is high and incur a fixed loss of 3 thousand BRL if demand is low.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1**:** Standard oven	A_2**:** Industrial oven
States of nature	S_1**:** High demand	S_2**:** Low demand
Uncertain event	Actual customer demand after the oven decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand BRL for each alternative–state pair.	

C2 - Payoff table [3]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Standard oven; A_2: Industrial oven	Alternative labels, in row order.
S_j	S_1: High demand; S_2: Low demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–high: 16 batches; Standard–low: 11 batches; Industrial–high: 13 batches; Industrial–low: 9 batches	Expected completed production batches for each oven under each demand condition.
$v^A(A_i)$	Standard oven: 3 thousand BRL/batch; Industrial oven: 5 thousand BRL/batch	Earnings per completed production batch from the chosen oven.
$f^S(S_j)$	High demand: 6 thousand BRL; Low demand: -3 thousand BRL	Fixed sales bonus or loss associated with the demand condition.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times v^A(A_1)$ $+ f^S(S_1)$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2) \times v^A(A_1)$ $+ f^S(S_2)$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1) \times v^A(A_2)$ $+ f^S(S_1)$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2) \times v^A(A_2)$ $+ f^S(S_2)$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	16×3 $+ 6$ $= 54$	11×3 $- 3$ $= 30$
A_2	13×5 $+ 6$ $= 71$	9×5 $- 3$ $= 42$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in thousand BRL).

Alternative	High demand (S_1)	Low demand (S_2)
Standard oven (A_1)	54	30
Industrial oven (A_2)	71	42

3 Urban Meal Subscription Service [C1-2 C2-5]

An urban meal subscription company must choose between a basic meal plan and a premium meal plan before customer interest is known. Interest may be strong or weak. The basic plan generates 4 thousand BRL per active subscriber and also provides a fixed operating contribution of 6 thousand BRL. The premium plan generates 6 thousand BRL per active subscriber and provides a fixed operating contribution of 9 thousand BRL. Strong interest adds another 2 thousand BRL per active subscriber to either plan, while weak interest adds 1 thousand BRL per active subscriber. The basic plan is expected to have 13 active subscribers under strong interest and 8 under weak interest; the premium plan is expected to have 10 active subscribers under strong interest and 6 under weak interest.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**
C2 Construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	A_1: Basic meal plan	A_2: Premium meal plan
States of nature	S_1: Strong customer interest	S_2: Weak customer interest
Uncertain event	Actual customer interest after the meal plan is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand BRL for each alternative–state pair.	

C2 - Payoff table [5]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Basic meal plan; A_2: Premium meal plan	Alternative labels, in row order.
S_j	S_1: Strong customer interest; S_2: Weak customer interest	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Basic–strong: 13 subscribers; Basic–weak: 8 subscribers; Premium–strong: 10 subscribers; Premium–weak: 6 subscribers	Expected active subscribers for each meal plan under each level of customer interest.
$v^A(A_i)$	Basic plan: 4 thousand BRL/subscriber; Premium plan: 6 thousand BRL/subscriber	Contribution per active subscriber from the chosen meal plan.
$v^S(S_j)$	Strong interest: 2 thousand BRL/subscriber; Weak interest: 1 thousand BRL/subscriber	Additional contribution per active subscriber from customer interest.
$f^A(A_i)$	Basic plan: 6 thousand BRL; Premium plan: 9 thousand BRL	Fixed operating contribution from the chosen meal plan.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v^A(A_1)+v^S(S_1)]$ $+f^A(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v^A(A_1)+v^S(S_2)]$ $+f^A(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v^A(A_2)+v^S(S_1)]$ $+f^A(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v^A(A_2)+v^S(S_2)]$ $+f^A(A_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$13\times$ $[4+2]$ $+6$ $=84$	$8\times$ $[4+1]$ $+6$ $=46$
A_2	$10\times$ $[6+2]$ $+9$ $=89$	$6\times$ $[6+1]$ $+9$ $=51$

Final payoff table (profit in thousand BRL).

Alternative	Strong customer interest (S_1)	Weak customer interest (S_2)
Basic plan (A_1)	84	46
Premium plan (A_2)	89	51

4 Regional Bookstore Expansion [C1-1 C3-2]

A regional bookstore must decide whether to open a children's section or maintain its current product selection before customer traffic is known. Traffic may be strong or weak. All values are in million BRL. Opening the children's section produces revenue of 72 under strong traffic and 35 under weak traffic, while maintaining the current selection produces revenue of 61 under strong traffic and 47 under weak traffic. The corresponding costs are 24 and 22 for the children's section under strong and weak traffic, respectively, and 25 and 23 for the current selection under strong and weak traffic, respectively.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Open children's section	A_2: Maintain current selection
States of nature	S_1: Strong traffic	S_2: Weak traffic
Uncertain event	Actual customer traffic after the product selection decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in million BRL for each alternative–state pair.	

C3 - Calculating profit and building the payoff table [2]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Open children's section; A_2: Maintain current selection	Alternative labels, in row order.
S_j	S_1: Strong traffic; S_2: Weak traffic	State labels, in column order; revealed after the choice.
$R(A_i, S_j)$	Children's section–strong: 72 million BRL; Children's section–weak: 35 million BRL; Current selection–strong: 61 million BRL; Current selection–weak: 47 million BRL	Total revenue generated by each product selection under each traffic condition.
$C(A_i, S_j)$	Children's section–strong: 24 million BRL; Children's section–weak: 22 million BRL; Current selection–strong: 25 million BRL; Current selection–weak: 23 million BRL	Total cost associated with each product selection under each traffic condition.

Profit equation.

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1) - C(A_1,S_1) = P(A_1,S_1)$	$R(A_1,S_2) - C(A_1,S_2) = P(A_1,S_2)$
A_2	$R(A_2,S_1) - C(A_2,S_1) = P(A_2,S_1)$	$R(A_2,S_2) - C(A_2,S_2) = P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$72-24 = 48$	$35-22 = 13$
A_2	$61-25 = 36$	$47-23 = 24$

Final payoff table (profit in million BRL).

Alternative	Strong traffic (S_1)	Weak traffic (S_2)
Open children's section (A_1)	48	13
Maintain current selection (A_2)	36	24

5 Greenhouse Crop Production [C1-1 C3-3]

A greenhouse farm is planning its production for an upcoming growing season. Revenue per production lot is 18 thousand EUR for tomatoes and 15 thousand EUR for peppers. Under favourable weather conditions, the farm expects to produce 8 lots of tomatoes and 7 lots of peppers, while under unfavourable weather conditions it expects to produce 5 lots of tomatoes and 4 lots of peppers. The cost per production lot is 7 thousand EUR under favourable weather and 10 thousand EUR under unfavourable weather.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	A_1: Produce tomatoes	A_2: Produce peppers
States of nature	S_1: Favourable weather	S_2: Unfavourable weather
Uncertain event	Weather conditions revealed after the production decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand EUR for each crop–weather-state pair.	

C3 - Profit and payoff table [3]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Produce tomatoes; A_2: Produce peppers	Alternative labels, in row order.
S_j	S_1: Favourable weather; S_2: Unfavourable weather	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Tomatoes–favourable: 8; Tomatoes–unfavourable: 5; Peppers–favourable: 7; Peppers–unfavourable: 4	Production lots for each crop under each weather condition.
$r(A_i)$	Tomatoes: 18 thousand EUR/lot; Peppers: 15 thousand EUR/lot	Revenue per production lot.
$c(S_j)$	Favourable weather: 7 thousand EUR/lot; Unfavourable weather: 10 thousand EUR/lot	Cost per production lot.

Profit equation.

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P(A_i, S_j) = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times r(A_1) = R(A_1, S_1)$	$q(A_1, S_2) \times r(A_1) = R(A_1, S_2)$
A_2	$q(A_2, S_1) \times r(A_2) = R(A_2, S_1)$	$q(A_2, S_2) \times r(A_2) = R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8 \times 18 = 144$	$5 \times 18 = 90$
A_2	$7 \times 15 = 105$	$4 \times 15 = 60$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times c(S_1) = C(A_1, S_1)$	$q(A_1, S_2) \times c(S_2) = C(A_1, S_2)$
A_2	$q(A_2, S_1) \times c(S_1) = C(A_2, S_1)$	$q(A_2, S_2) \times c(S_2) = C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8 \times 7 = 56$	$5 \times 10 = 50$
A_2	$7 \times 7 = 49$	$4 \times 10 = 40$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1) = P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2) = P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1) = P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2) = P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$144 - 56 = 88$	$90 - 50 = 40$
A_2	$105 - 49 = 56$	$60 - 40 = 20$

Final payoff table (profit in thousand EUR).

Alternative	Favourable weather (S_1)	Unfavourable weather (S_2)
Produce tomatoes (A_1)	88	40
Produce peppers (A_2)	56	20

6 Fitness Center Membership Plans [C1-2 C3-5]

A fitness center must choose between a basic membership plan and a premium membership plan before monthly demand is known. Demand may be high or low. Quantities, in hundreds of memberships, are 9 and 6 for the basic plan and 7 and 4 for the premium plan under high and low demand, respectively. The plan-based revenue rates are 8 and 13 thousand USD per hundred memberships for the basic and premium plans. High demand adds a state-based revenue rate of 2 thousand USD per hundred memberships; low demand adds 1. The plan-based cost rates are 3 and 6 thousand USD per hundred memberships, respectively. The fixed operating costs are 12 thousand USD under high demand and 16 thousand USD under low demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**

C3 Use revenue and costs to construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	A_1: Offer a basic membership plan	A_2: Offer a premium membership plan
States of nature	S_1: High monthly demand	S_2: Low monthly demand
Uncertain event	Monthly market conditions revealed after the membership-plan decision	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand USD for each plan–market-state pair.	

C3 - Profit and payoff table

[5]

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Offer a basic membership plan; A_2: Offer a premium membership plan	Alternative labels, in row order.
S_j	S_1: High monthly demand; S_2: Low monthly demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Basic–high: 9; Basic–low: 6; Premium–high: 7; Premium–low: 4 hundreds of memberships	Membership lots for each plan under each demand level.
$r^A(A_i)$	Basic plan: 8 thousand USD/100 memberships; Premium plan: 13 thousand USD/100 memberships	Plan-based revenue rate per hundred memberships.
$r^S(S_j)$	High demand: 2 thousand USD/100 memberships; Low demand: 1 thousand USD/100 memberships	Additional revenue rate per hundred memberships from monthly demand.
$c^A(A_i)$	Basic plan: 3 thousand USD/100 memberships; Premium plan: 6 thousand USD/100 memberships	Plan-based cost rate per hundred memberships.
$f_c(S_j)$	High demand: 12 thousand USD; Low demand: 16 thousand USD	Fixed operating cost under each demand condition.

Profit equation.

$$R = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right], \qquad C = q(A_i, S_j)c^A(A_i) + f_c(S_j), \qquad P(A_i, S_j) = R - C.$$

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times [r^A(A_1)+r^S(S_1)] = R(A_1, S_1)$	$q(A_1, S_2)\times [r^A(A_1)+r^S(S_2)] = R(A_1, S_2)$
A_2	$q(A_2, S_1)\times [r^A(A_2)+r^S(S_1)] = R(A_2, S_1)$	$q(A_2, S_2)\times [r^A(A_2)+r^S(S_2)] = R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$9\times [8+2] = 90$	$6\times [8+1] = 54$
A_2	$7\times [13+2] = 105$	$4\times [13+1] = 56$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1)\times c^A(A_1) + f_c(S_1) = C(A_1, S_1)$	$q(A_1, S_2)\times c^A(A_1) + f_c(S_2) = C(A_1, S_2)$
A_2	$q(A_2, S_1)\times c^A(A_2) + f_c(S_1) = C(A_2, S_1)$	$q(A_2, S_2)\times c^A(A_2) + f_c(S_2) = C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$9\times 3 + 12 = 39$	$6\times 3 + 16 = 34$
A_2	$7\times 6 + 12 = 54$	$4\times 6 + 16 = 40$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$90-39$ $=51$	$54-34$ $=20$
A_2	$105-54$ $=51$	$56-40$ $=16$

Final payoff table (profit in thousand USD).

Alternative	High monthly demand (S_1)	Low monthly demand (S_2)
Basic membership plan (A_1)	51	20
Premium membership plan (A_2)	51	16

7 Restaurant Expansion Strategy [C4-2]

A restaurant company must choose a strategy for expanding into a new location. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Demand		
Alternative	High	Medium	Low
Full-service restaurant	78	35	8
Casual restaurant	72	48	32
Express restaurant	95	42	28
Postpone expansion	25	25	25

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin [2]



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_3$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_2$

Maximax

$$\begin{aligned} M_1 &= \max\{78, 35, 8\} = 78 \\ M_2 &= \max\{72, 48, 32\} = 72 \\ M_3 &= \max\{95, 42, 28\} = 95 \\ M_4 &= \max\{25, 25, 25\} = 25 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{78, 35, 8\} = 8 \\ m_2 &= \min\{72, 48, 32\} = 32 \\ m_3 &= \min\{95, 42, 28\} = 28 \\ m_4 &= \min\{25, 25, 25\} = 25 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	78	72	95	25	$A^*_{\text{Maximax}} = A_3$
Min	8	32	28	25	$A^*_{\text{Maximin}} = A_2$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_3 \quad \text{(Express restaurant)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_2 \quad \text{(Casual restaurant).}$$

8 Hotel Development Strategy [C4-2]

A hotel company must choose a strategy for developing a new property. Tourist demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Tourist demand		
Alternative	High	Medium	Low
Large hotel	72	34	10
Medium hotel	65	46	22
Small hotel	95	44	31
No development	58	40	35

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**

Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_3$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{72, 34, 10\} = 72 \\ M_2 &= \max\{65, 46, 22\} = 65 \\ M_3 &= \max\{95, 44, 31\} = 95 \\ M_4 &= \max\{58, 40, 35\} = 58 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{72, 34, 10\} = 10 \\ m_2 &= \min\{65, 46, 22\} = 22 \\ m_3 &= \min\{95, 44, 31\} = 31 \\ m_4 &= \min\{58, 40, 35\} = 35 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	72	65	95	58	$A^*_{\text{Maximax}} = A_3$
Min	10	22	31	35	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_3 \quad \text{(Small hotel)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(No development)}.$$

9 Agricultural Technology Investment Strategy **[C4-2]**

An agricultural company must choose a strategy for introducing a new automated farming system. Market conditions could be highly favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

	Market conditions			
Alternative	Highly favourable	Favourable	Stable	Unfavourable
Full automation	125	58	8	-42
Partial automation	88	64	30	4
Basic automation	62	52	38	20
Equipment leasing	48	43	35	28

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{125, 58, 8, -42\} = 125 \\ M_2 &= \max\{88, 64, 30, 4\} = 88 \\ M_3 &= \max\{62, 52, 38, 20\} = 62 \\ M_4 &= \max\{48, 43, 35, 28\} = 48 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{125, 58, 8, -42\} = -42 \\ m_2 &= \min\{88, 64, 30, 4\} = 4 \\ m_3 &= \min\{62, 52, 38, 20\} = 20 \\ m_4 &= \min\{48, 43, 35, 28\} = 28 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	125	88	62	48	$A^*_{\text{Maximax}} = A_1$
Min	-42	4	20	28	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Full automation)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(Equipment leasing).}$$

10 Summarized Solutions [C1-C4]

10.1 Local Food Market Stall C1–1 C2–2

Element	Description
Alt.	**A1:** Standard stall **A2:** Deluxe stall
States	**S1:** High demand **S2:** Low demand
Events	Actual market demand after the stall decision
Cons.	Profit in thousand US dollars for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	18	11
$\mathbf{A_2}$	31	6

10.2 Small Bakery Production C1–1 C2–3

Element	Description
Alt.	**A1:** Standard oven **A2:** Industrial oven
States	**S1:** High demand **S2:** Low demand
Events	Actual customer demand after the oven decision
Cons.	Profit in thousand BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$16(3)+6 = 54$	$11(3)-3 = 30$
$\mathbf{A_2}$	$13(5)+6 = 71$	$9(5)-3 = 42$

10.3 Urban Meal Subscription Service C1–2 C2–5

Element	Description
Alt.	**A1:** Basic meal plan **A2:** Premium meal plan
States	**S1:** Strong customer interest **S2:** Weak customer interest
Events	Actual customer interest after the meal plan is selected
Cons.	Profit in thousand BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$13(4+2)+6 = 84$	$8(4+1)+6 = 46$
$\mathbf{A_2}$	$10(6+2)+9 = 89$	$6(6+1)+9 = 51$

10.4 Regional Bookstore Expansion C1–1 C3–2

Element	Description
Alt.	**A1:** Open children's section **A2:** Maintain current selection
States	**S1:** Strong traffic **S2:** Weak traffic
Events	Actual customer traffic after the product selection decision
Cons.	Profit in million BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$72-24 = 48$	$35-22 = 13$
$\mathbf{A_2}$	$61-25 = 36$	$47-23 = 24$

10.5 Greenhouse Crop Production C1–1 C3–3

Element	Description
Alt.	**A1:** Produce tomatoes **A2:** Produce peppers
States	**S1:** Favourable weather **S2:** Unfavourable weather
Events	Weather conditions revealed after the production decision
Cons.	Profit in thousand EUR for each crop–weather-state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$8(18) - 8(7) = 88$	$5(18) - 5(10) = 40$
$\mathbf{A_2}$	$7(15) - 7(7) = 56$	$4(15) - 4(10) = 20$

10.6 Fitness Center Membership Plans C1–2 C3–5

Element	Description
Alt.	**A1:** Offer a basic membership plan **A2:** Offer a premium membership plan
States	**S1:** High monthly demand **S2:** Low monthly demand
Events	Monthly market conditions revealed after the membership-plan decision
Cons.	Profit in thousand USD for each plan–market-state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$9(8+2) - [9(3)+12] = 51$	$6(8+1) - [6(3)+16] = 20$
$\mathbf{A_2}$	$7(13+2) - [7(6)+12] = 51$	$4(13+1) - [4(6)+16] = 16$

10.7 Restaurant Expansion Strategy C4–2

	Max	Min
A_1	78	8
A_2	72	32
A_3	95	28
A_4	25	25
A^*_{Maximax}	A_3	
A^*_{Maximin}		A_2

10.8 Hotel Development Strategy C4–2

	Max	Min
A_1	72	10
A_2	65	22
A_3	95	31
A_4	58	35
A^*_{Maximax}	A_3	
A^*_{Maximin}		A_4

10.9 Agricultural Technology Investment Strategy C4–2

	Max	Min
A_1	125	-42
A_2	88	4
A_3	62	20
A_4	48	28
A^*_{Maximax}	A_1	
A^*_{Maximin}		A_4